Marstek Device Local API（Rev 3.1.2）

The Local API is provided “as is” for local use only.Use at your own risk. Marstek is not liable for any damages, data loss, or legal issues caused by your use of the API.You are responsible for lawful and appropriate use.

Ⅰ. Preface：

Welcome!

This document provides an introduction to the Local API for Marstek devices, which is available to device owners and enables integration with third-party systems.

While Marstek offers an official mobile app and cloud services, this Local API is designed for advanced users who wish to gain greater control over their devices and seamlessly integrate them into other management platforms.

Ⅱ. General Description

Marstek devices communicate with third-party systems over a Local Area Network (LAN). Before using this API, please ensure that:

1. The Marstek device is properly connected to your home network.
2. The Local API feature has been enabled via the Marstek mobile app

Please note that different Marstek models may support only a subset of the commands described in this documentation. Additionally, enabling the Local API may cause certain built-in features of the device to be disabled in order to prevent command conflicts. For detailed information about supported commands and any functional limitations for specific models, please refer to **Chapter 4** of this document.

2.1 Protocol Format

The protocol utilizes the JSON format, with commands primarily categorized into query commands, configuration commands, and certain special commands.

Command Format

Property	Type	Description
id	number or string	An identifier established by the Client
method	string	A Structured value that holds the parameter values to be used during the invocation of the method.
params	object	Parameters that the method takes.

Example

```
{
    "id": 0,
    "method": "string",
    "params": {
    "id":0
    }
}
```

Device Response Format

Property	Type	Description
id	number or string	Identifier of this request.
src	string	Name of the source of the request.
result	object	Parameters that the method takes.

Error code

Code	MessageMeaning	
-32700	Parse error	Invalid JSON was received by the server.An error occurred on the server while parsing the JSON text
-32600	Invalid Request	The JSON sent is not a valid Request object
-32601	Method not found	The method does not exist / is not available.
-32602	Invalid params	Invalid method parameter(s).
-32603	Internal error	Internal JSON-RPC error
-32000 to -32099	Server error	Reserved for implementation-defined server-errors

The remainder of the space is available for application defined errors.

Example response on success

```
{
    "id": 0,
    "src": "device",
    "result": {
    "id":0
    }
}
```

Example response on error

```
{
    "id": 0,
    "src": "Venus-24215ee580e7",
    "error": {
    "code": -32700,
    "message": "Parse error",
    "data": 402
    }
}
```

2.2 API over UDP

2.2.1 First-Time Use

When users first use the Local API service, they need to follow the configuration process below:

1. Connect the device to power and turn it on;
2. Use the Marstek APP to connect and bind the device, and configure the WiFi network for thedevice or connect it to the Ethernet;
3. Enable the device's API feature in the APP, and set the UDP port number. The default port number is 30000, and the recommended port number is between 49152 and 65535

After completing the above operations, the Marstek device can normally receive UDP commands from the same local area network.

2.2.2 Discovering Devices

To discover Marstek devices within the LAN, a UDP broadcast is utilized. The broadcast content is as follows:

```
{
    "id": 0,
    "method": "Marstek.GetDevice",
    "params": {
    "ble_mac":"0"
    }
}
```

If there are Marstek devices within the LAN, taking Venus C as an example, the following response will be received

```
{
    "id": 0,
    "src": "VenusC-123456789012",
    "result": {
    "device":"VenusC",
    "ver":111,
    "ble_mac":"123456789012",
    "wifi_mac":"123456789012",
    "wifi_name":"MY_HOME",
    "ip":"192.168.1.11"
    }
}
```

The device's IP address can be directly obtained from the Marstek APP or the home router. If this functionality is to be used on a long-term basis, it is recommended to configure the device with a static* IP address.*

III. Components

This chapter mainly introduces the components and services supported by the Marstek device.

3.1 Marstek

Marstek contains some basic information about the product, and is mainly used for discovering devices and querying basic device information.

Marstek.GetDevice: Locate Marstek devices on the local area network.

3.1.1 Marstek.GetDevice

Sending:

```
{
    "id": 0,
    "method": "Marstek.GetDevice",
    "params": {
    "ble_mac":"123456789012"
    }
}
```

Response:

```
    {
        "id": 0,
        "src": "VenusC-123456789012",
        "result": {
        "device":"VenusC",
        "ver":111,
        "ble_mac":"123456789012",
        "wifi_mac":"012123456789",
        "wifi_name": "MY_HOME",
        "ip":"192.168.1.11"
        }
    }
```

3.2 WiFi

The WiFi component is mainly used for configuring the device's WiFi and obtaining the device's basic network information.

Wifi.GetStatus: Obtain the device's basic network information.

3.2.1 Wifi.GetStatus

Sending:

Property (params)	Type	Description
id	*number*	ID of Instance

Response:

Property (result)	Type	Description
id	*number*	ID of Instance
wifi_mac	*string*	WiFi MAC
ssid	*string or null*	WiFi name
rssi	*number*	WiFi signal strength
sta_ip	*string or null*	Device IP
sta_gate	*string or null*	Gateway
sta_mask	*string or null*	Subnet mask
sta_dns	*string or null*	DNS

Example:

Sending:

```
{
    "id": 1,
    "method": "Wifi.GetStatus",
    "params": {
    "id": 0
    }
}
```

Response:

```
{
    "id": 1,
    "src": "VenusC-mac",
    "result": {
    "id": 0,
    "wifi_mac": "620b0c877705"
    "ssid":"Hame",
    "rssi": -59,
    "sta_ip":"192.168.137.41",
    "sta_gate":"192.168.137.1",
    "sta_mask":"255.255.255.0",
    "sta_dns":"192.168.137.1"
    }
}
```

3.3 Bluetooth

The BLE (Bluetooth) component can view the Bluetooth-related information of the device

BLE.GetStatus: Check the Bluetooth connection status of the device.

3.3.1 BLE.GetStatus

Sending:

Property (params)	Type	Description
id	*number*	ID of Instance

Response:

Property (result)	Type	Description
state	*string*	Bluetooth state
ble_mac	*string*	Bluetooth MAC

Example

Sending:

```
{
    "id": 1,
    "method": "BLE.GetStatus",
    "params": {
    "id": 0
    }
}
```

Response:

```
{
    "id": 1,
    "src": "VenusC-mac",
    "result": {
    "id": 0,
    "state":"connect",
    "ble_mac":"50cf14640fac"
    }
}
```

3.4 Battery

The Bat (Battery) component contains basic information about the device's battery

Bat.**GetStatus**: Query the device's battery information and operating status.

3.4.1 Bat.GetStatus

Sending:

Property (params)	Type	Description
id	*number*	ID of Instance

Response:

property (result)	Type	Description
id	*number*	ID of Instance
soc	*string*	soc
charg_flag	*boolean*	Charging permission flag
dischrg_flag	*boolean*	Discharge permission flag
bat_temp	*number or null*	Battery temperature, [°C]
bat_capacity	*number or null*	Battery remaining capacity, [Wh]
rated_capacity	*number or null*	Battery rated capacity, [Wh]

Example

Sending:

```
{
    "id": 1,
    "method": "Bat.GetStatus",
    "params": {
    "id": 0
    }
}
```

Response:

```
{
    "id": 1,
    "src": "VenusC-mac",
    "result": {
    "id": 0,
    "soc": 90,
    "charg_flag": true,
    "dischrg_flag": true,
    "bat_temp": 25.0,
    "bat_capacity": 256.0,
    "rated_capacity": 2560.0
    }
}
```

3.5 PV(only for Venus A/D)

The PV (Photovoltaic) component contains the photovoltaic information connected to thedevice

PV.**GetStatus**: Query the device's connected photovoltaic information and power generationstatus

3.5.1 PV.GetStatus

Sending:

Property (params)	Type	Description
id	*number*	ID of Instance

Response:

Property (result)	Type	Description
id	*number*	ID of Instance
pv_power	*number*	Photovoltaic charging power, [W]
pv_voltage	*number*	Photovoltaic charging voltage, [V]

Property (result)	Type	Description
pv_current	*number*	Photovoltaic charging current, [A]
PV_state	*number*	Photovoltaic status 1:Work 0:Standby

Example:

```
    {
        "id": 1,
        "method": "PV.GetStatus",
        "params": {
        "id": 0
        }
    }
```

Response:

```
    {
        "id": 1,
        "src": "VenusC-mac",
        "result": {
        "id": 0,
        "pv1_power": 0,
        "pv1_voltage": 10,
        "pv1_current": 0,
        "pv1_state": 0,
        "pv2_power": 0,
        "pv2_voltage": 9,
        "pv2_current": 0,
        "pv2_state": 0,
        "pv3_power": 0,
        "pv3_voltage": 10,
        "pv3_current": 0,
        "pv3_state": 0,
        "pv4_power": 0,
        "pv4_voltage": 10,
        "pv4_current": 0,
        "pv4_state": 0
        }
    }
```

3.6 ES

The ES (Energy System) component contains the device's basic power information and energy statistics, and can configure or monitor the device's operating status.

1. ES.GetStatus: Query the device's basic electrical energy information.
2. ES.SetMode: Configure the device's operating mode.
3. ES.GetMode: Get information about the operating mode of the device.

3.6.1 ES.GetStatus

Sending:

Property (params)	Type	Description
id	*number*	ID of Instance

Response:

Property (result)	Type	Description
id	number or *null*	ID of Instance
bat_soc	number or *null*	Total battery SOC，[%]
bat_cap	number or *null*	Total battery capacity, [Wh]
pv_power	number or *null*	Solar charging power, [W]
ongrid_power	number or *null*	Grid-tied power, [W]
offgrid_power	number or *null*	Off-grid power, [W]
bat_power	number or *null*	Battery power, [W]
total_pv_energy	number or *null*	Total solar energy generated, [0.01*KWh]
total_grid_output_energy	number or *null*	Total grid output energy, [Wh]
total_grid_input_energy	number or *null*	Total grid input energy, [Wh]
total_load_energy	number or *null*	Total load (or off-grid) energy consumed,[Wh]

Example：

Sending:

```
{
    "id": 1,
    "method": "ES.GetStatus",
    "params": {
    "id": 0
    }
}
```

Response:

```
{
    "id": 1,
    "src": "VenusE-24215edb178f",
    "result": {
    "id": 0,
```

```
        "bat_soc": 98,
        "bat_cap": 5120,
        "pv_power": 0,
        "ongrid_power": 100,
        "offgrid_power": 0,
        "total_pv_energy": 0,
        "total_grid_output_energy": 2548,
        "total_grid_input_energy": 3273,
        "total_load_energy": 0
        }
    }
```

3.6.2 ES.SetMode

Sending:

Property (params)	Type	Description
id	number	ID of Instance
config	object	Config Parameters

Object: config

Property (config)	Type	Description
mode	*string*	Device power generation mode, including the following : "Auto"; "AI"; "Manual"; "Passive"; "Ups".
auto_cfg	*object*	Configuration parameters for Auto mode
ai_cfg	*object*	Configuration parameters for AI mode
manual_cfg	*object*	Configuration parameters for Manual mode
passive_cfg	*object*	Configuration parameters for Passive mode
ups_cfg	*object*	Configuration parameters for Ups mode

Object: auto_cfg

Property (auto_cfg)	Type	Description
enable	number	ON: 1; OFF: Set another mode

Object: ai_cfg

Property (ai_cfg)	Type	Description
enable	number	ON: 1; OFF: Set another mode

Object: manual_cfg

Property (manual_cfg)	Type	Description
time_num	number	Time period serial number, Venus A/C/D/E supports 0-9; Venus E mini supports 0-5
start_time	*string*	Start time, hours: minutes, [hh:mm]
end_time	*string*	End time, hours: minutes, [hh:mm]
week_set	number	Week, a byte 8 bits, the low 7 bits effective, the highest nvalid, 0000 0001 (1) on behalf of Monday open, 0000 0011 (3) on behalf of Monday and Tuesday open,
power	number	Setting power,[W]
enable	number	ON: 1; OFF: 0
manual_set (**Only for Venus E mini**)	number	0: disable; 1: enable charge; 2: enable discharge; 3: auto

Object: passive_cfg

Property (passive_cfg)	Type	Description
power	number	Setting power,[W]
cd_time	number	Power countdown,[s]

Object: ups_cfg

Property (ups_cfg)	Type	Description
enable	number	ON: 1; OFF: 0

Response：

Property（result）	Type	Description
id	number	ID of Instance
set_result	boolean	"true":succeeded in setting; "false":failed in setting

Example

Sending:

```
    /* Auto Mode Example */
    {
        "id": 1,
```

```
      "method": "ES.SetMode",
          "params": {
          "id": 0,
          "config": {
              "mode": "Auto",
              "auto_cfg": {
              "enable": 1
              }
          }
      }
  }

/* AI Mode Example */
  {
      "id": 1,
      "method": "ES.SetMode",
      "params": {
      "id": 0,
          "config": {
          "mode": "AI",
              "ai_cfg": {
              "enable": 1
              }
          }
      }
  }
/* Manual Mode Example */
  {
      "id": 1,
      "method": "ES.SetMode",
          "params": {
          "id": 0,
              "config": {
              "mode": "Manual",
                  "manual_cfg": {
                  "time_num": 1,
                  "start_time": "08:30",
                  "end_time": "20:30",
                  "week_set": 127,
                  "power": 100,
                  "enable": 1,
                  "manual_set": 1
                  }
              }
          }
  }

  /* Passive Pattern Example */
  {
  "id": 1,
  "method": "ES.SetMode",
      "params": {
      "id": 0,
```

```
            "config": {
            "mode": "Passive",
                "passive_cfg": {
                "power": 100,
                "cd_time": 300
                }
            }
        }
    }

    /* UPS Mode Example */
    {
    "id": 1,
    "method": "ES.SetMode",
        "params": {
        "id": 0,
            "config": {
            "mode": "UPS",
                "ups_cfg": {
                "enable": 1
                }
            }
        }
    }
```

Response:

```
    {
        "id": 1,
        "src": "Venus-mac",
        "result": {
        "id": 0,
        "set_result": ture
        }
    }
```

3.6.3 ES.GetMode

Sending:

Property (params)	Type	Description
id	*number or null*	ID of Instance

Response:

Property (result)	Type	Description
id	*number or null*	ID of Instance

Property (result)	Type	Description
mode	*number or null*	Auto：Auto mode; AI: AI mode; Manual: manual mode; Passive: Passive control mode
ongrid_power	*number or null*	Grid-tied power, [W]
offgrid_power	*number or null*	Off-grid power, [W]
bat_soc	*number or null*	SOC，[%]
ct_state	*number or null*	CT Status，0: Not connected; 1: Connected. Note: Effective in Auto mode and AI mode.
a_power	*number or null*	Phase A Power, [W] Note: Effective in Auto mode and AI mode
b_power	*number or null*	Phase B Power, [W] Note: Effective in Auto mode and AI mode
c_power	*number* *or null	Phase C Power, [W] Note: Effective in Auto mode and AI mode
total_power	*number or null*	CT total power, [W] Note: Effective in automatic mode and AI mode
input_energy	*number or null*	Cumulative Input Energy [Wh] （*0.1）Note: Effective in automatic mode and AI mode
output_energy	*number or null*	Total cumulative output energy [Wh] （*0.1）Note: Effective in automatic mode and AI mode

Example

Sending:

```
{
    "id": 1,
    "method": "ES.GetMode",
    "params": {
    "id": 0
    }
}
```

Response:

```
{
    "id": 1,
    "src": "VenusE-24215edb178f",
```

```
        "result": {
        "id": 0,
        "mode": "Auto",
        "ongrid_power": 100,
        "offgrid_power": 0,
        "bat_soc": 98,
        "ct_state": 0,
        "a_power": 0,
        "b_power": 0,
        "c_power": 0,
        "total_power": 0
        "input_energy": 3086320,
        "output_energy": 4487510
        }
    }
```

3.7 EM

The Energy Meter (EM) module contains status information and power measurement data from the energy meter, or data obtained from the current transformer (CT).

EM.GetStatus: Queries the basic status and data information of the energy meter.

3.7.1 EM.GetStatus

Sending:

Property (params)	Type	Description
id	*number or null*	ID of Instance

Response:

Property (result)	Type	Description
id	*number or null*	ID of Instance
ct_state	*number or null*	CT (Current Transformer) status: 0: Not connected 1: Connected
a_power	*number or* *null	Phase A power,[W]
b_power	*number or null*	Phase B power,[W]
c_power	*number or null*	Phase C power,[W]
total_power	*number or null*	Total power,[W]
input_energy	*number or null*	Cumulative Input Energy [0.1 Wh]
output_energy	*number or null*	Total cumulative output energy [0.1 Wh]

Sending:

```
{
    "id": 1,
    "method": "EM.GetStatus",
    "params": {
    "id": 0
    }
}
```

Response:

```
{
    "id": 1,
    "src": "VenusD-009b08a5ac28",
    "result": {
    "id": 0,
    "ct_state": 1,
    "a_power": 0,
    "b_power": 0,
    "c_power": 0,
    "total_power": 0,
    "input_energy": 0,
    "output_energy": 0
    }
}
```

3.8 SYS

Includes some common system control commands, like DOD and Bluetooth lock features

3.8.1 DOD

This chapter contains the DOD configuration instructions. The device's default DOD value is 88, with a setting range of 30-88.

Sending:

Property (params)	Type	Description
id	*number or null*	ID of Instance
Value	number	DOD Value (range 30-88)

Response：

Property（result）	Type	Description
id	number	ID of Instance
set_result	boolean	true：Success false：fai

Sending:

```
{
    "id": 0,
    "method": "DOD.SET",
    "params": {
    "value": 36
    }
}
```

Response：

```
{
    "id": 1,
    "src": "Venus-mac",
    "result": {
    "id": 0,
    "set_result": ture
    }
}
```

3.8.2 Ble_block

This chapter covers the Bluetooth lock feature, including turning Bluetooth broadcasting on and off

Sending:

Property (params)	Type	Description
id	*number or null*	ID of Instance
enable	number	0：enable 1：disable

Response：

Property（result）	Type	Description
id	number	ID of Instance
set_result	boolean	true：Success false：fai

Sending:

```
{
    "id": 0,
    "method": "Ble.Adv",
    "params": {
    "enable": 0
    }
}
```

Response：

```
{
    "id": 1,
    "src": "Venus-mac",
    "result": {
    "id": 0,
    "set_result": ture
    }
}
```

3.8.3 Led_Ctrl

This chapter covers the LED functions of the switchgear panel:

Sending:

Property (params)	Type	Description
id	*number or null*	ID of Instance
state	number	1：on 0：off

Sending:

```
{
    "id": 0,
    "method": "Led.Ctrl",
    "params": {
    "state": 0
    }
}
```

Response:

```
{
    "id": 1,
    "src": "Venus-mac",
    "result": {
    "id": 0,
    "set_result": ture
    }
}
```

3.8.4 Version_Set

This chapter includes settings for charge and discharge power version information

Property (params)	Type	Description
id	*number or null*	ID of Instance
version	number	800W, 1200W(VA), 1500W（VA）,2200W(VD)，and2500W(VE)。

Example

Sending:

```
{
  "id": 0,
  "method": "Set.Ver",
      "params": {
        "version": 2500
      }
}
```

Response:

```
{
    "id": 1,
    "src": "Venus-mac",
    "result": {
        "id": 0,
        "set_result": ture
    }
}
```

3.8.5 Factory_Reset

This chapter includes information on resetting to factory settings

Property (params)	Type	Description
id	*number or null*	ID of Instance
type	number	1. Reset to factory settings, clear accumulated data 2. Reset to factory settings, keep accumulated data

Example

Sending:

```
{
  "id": 0,
  "method": "Reset.Factory",
      "params": {
        "type": 1
      }
}
```

Response:

```
{
    "id": 1,
    "src": "Venus-mac",
    "result": {
        "id": 0,
        "set_result": ture
    }
}
```

IV . Devices

This chapter will describe the extent of support for the components and services in this API documentation by different Marstek devices, as well as some proprietary information.

4.1 Venus C/E

1. Marstek
2. WiFi
3. Bluetooth
4. Battery
5. ES
6. EM
7. SYS（FW>=150 available）

4.2 Venus D/Venus A

1. Marstek
2. WiFi
3. Bluetooth
4. Battery
5. PV
6. ES
7. EM
8. SYS（FW>=150 available）

4.3 Venus E mini

1. Marstek
2. WiFi
3. Bluetooth
4. Battery
5. ES
6. EM

7. SYS（DOD、Ble_block、Led_Ctrl）